

4 Results

As stated in the Introduction the analysis of our regression results will follow the structure of Su and Yu, 2020. The first stage of this analysis will follow the general assessment of spatial autocorrelation within our model. This is necessary, since normally distributed residuals must be accounted for in case of spatial proximity. This will be examined in form of the Moran's I Test (Getis (2009)) and a LISA model (Çelebioğlu (2010), Su and Yu (2020)). A second-stage of our results analysis will include the model specification among the ordinary least squares estimation model and a range of spatial regression models. The third stage of the results section will display the concrete and significant results obtained by the chosen model.

4.1 Spatial Dependence

Steijn et al., 2022 have already highlighted the increasing importance of geographic proximity due to pooling of labor, technology and transportation. It is therefore essential to evaluate the impact of spatial proximity within our regression. As stated previously, the global Moran's I test is applied to accounted for spatial autocorrelation (Cliff and Ord (1970)). Table 1 shows that the Moran's I value is greater than 0 and significant which indicates a degree of spatial autocorrelation among the specialized industrial labor (*spec*) and the residuals of our initial OLS regression. This implies that a switch to spatial econometric analysis is necessary.

	<i>spec</i>	<i>ols_residuals</i>
Moran's I	0.039255	0.007000
Z-value	22.23	9.7747
p-value	2.2e-16	2.2e-16

Table 1: Spatial autocorrelation for *spec* and the residuals in OLS

Demonstrating the spatial dependence of industrial labor pooling is figure 2 which shows a Moran's I scatter plot displaying the average values of our dependent variable. This plot is dissected into four quadrants: high-high, low-high, low-low, high-low representing the first,

second, third and fourth quadrant respectively (Su and Yu (2020)). As can be seen in figure 2, the average values seem to concentrate towards the upper-right corner (first quadrant) creating a upwards sloping fitted line. This graphically confirms the results gained in table 1.

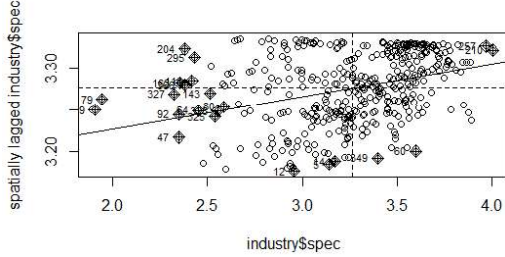


Figure 2: Moran's I scatterplot

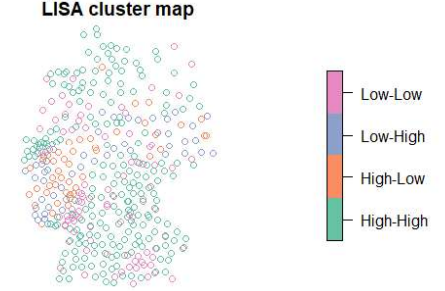


Figure 3: LISA cluster diagram

Furthermore the Moran's I values for significant regions are pooled together in figure 3 to give a geographic overview of spatially correlated regions when considering specialized industrial labor pools. Figure 3 also highlights the clustering of industries predominantly in the southern, central and the "Ruhr" area in western Germany.

4.2 Model Specification

After identifying spatial autocorrelation within our dataset, it is further necessary to specify the most suitable model for an accurate result analysis. Su and Yu, 2020 choose among two different categories of spatial regression models: spatial lag model (SLM, for this paper SLX) and the spatial error model (SEM). Additionally the Spatial Durbin Model (SDM) is considered as a subcategory of the SLM. These models differ from the ordinary OLS model by adding spatial weights to each regression coefficient.

$$\text{OLS: } y = X\beta + \varepsilon$$

$$\text{SLX: } y = X\beta + WX\theta + \varepsilon$$

$$\text{SEM: } y = X\beta + u \quad \text{where} \quad u = \lambda Wu + \varepsilon$$

$$\text{SDM: } y = \rho Wy + X\beta + WX\theta + \varepsilon$$

With W denoting the spatial matrix of distances added to the matrix of independent variables X and for the SDM dependent variable matrix Y . ρ , θ and λ represent spatial autoregressive coefficients (see Chi and Zhu (2008)). Su and Yu, 2020 use the LM-Test to select the appropriate model by investigating the significance of the LM-lag and LM-error test results. If the (robust) LM-lag is significant, the SLX model is chosen. If the (robust) LM-error is significant, the SEM model is chosen. This also applies in case of both LM-lag and LM-error

being significant. Following the procedure described previously the results of table 2 show, that the SLX model is preferred over the SEM model.

	<i>LM-lag</i> (robust LM-Lag)	<i>LM-error</i> (robust LM-error)
statistic	19.56 (16.755)	3.0264 (0.22204)
p-value	9.75e-06 (4.252e-05)	0.08192 (0.6375)
significance (under $\alpha = 0.05$)	yes	no

Table 2: LM-Test results

The next step would be to assess if the SDM model can be restricted to the SLX model. The more detailed SDM model also includes a spatial lag of the dependent variable (see Mur and Angulo (2006)). According to Su and Yu, 2020 a LR-Test (Likelihood Ratio Test) is best suited for this assessment. With a higher LR-ratio for the SDM (under p-value = 2.194e-05) it can be concluded that the SDM provides a better fit to our data as well as a rejection of H_0 that the SDM can be restricted to the SLX. The SDM will therefore be used as our regression model.

4.3 Results

The regression results (3) for the SDM give a compact overview of the influence of each variable on *spec* by including a non-spatial and spatially lagged result (see 4.2). Table 3 shows that, while some variables (*wage*, *car*, *gva*, *eastwest*) are significant on both levels, there are some important differentiations captured in the model outcome. *rent* and *tech* both became more significant when incorporating neighboring spatial spillovers. The first being in line with Dunse et al., 2005 who have found evidence of industrial rent clustering. The second effect can be related to Steijn et al., 2022 by demonstrating the importance of technological spillover effects in more technologically advanced areas. Furthermore *car* has become less significant but more impactful with it having the strongest impact of the spatially lagged variables. Under the assumption that *car* relates to an accessible and extensive road infrastructure (Yu and Zhao (2021)), we can confirm that transport infrastructure plays a crucial role in agglomeration economies (Arauzo-Carod et al. (2010)). A similar effect is obtained for *wage*. This also confirms the findings of Arauzo-Carod et al., 2010 by showcasing the spatial agglomeration of industrial wages. Interestingly, *skill* and *academic* lose significance when attributed with a spatial lag. This contradicts Arauzo-Carod et al., 2010 by weakening the assumption that there is a spatial agglomeration effect of industry in more educated areas.

Table 3: Comparison of Coefficients and Lagged Coefficients in SDM

	<i>Coefficient</i>	<i>Lagged Coefficient</i>
rent	-0.064 (0.060)	-5.386* (3.071)
skill	0.254* (0.140)	7.587 (18.253)
academic	-0.117*** (0.038)	-5.212 (4.406)
wage	0.242*** (0.075)	22.511*** (7.599)
car	0.505*** (0.115)	30.657** (15.408)
gva	0.698*** (0.025)	15.278*** (4.437)
tech	0.023 (0.021)	11.409*** (2.746)
eastwest	-0.132*** (0.037)	-8.528*** (3.250)
Constant	-386.707** (182.288)	
Observations	400	
Wald Test	15.042*** (df = 1)	
LR Test	16.820*** (df = 1)	
Note:	*p<0.1; **p<0.05; ***p<0.01	